Tracking, Uncertainty, and Information Over Time: A Study of the 2D Constant-Velocity Kalman Filter

Anonymous Submission
Assignment II

***Abstract*—This report implements and analyses a Kalman filter for two-dimensional object tracking under a constant-velocity motion model. Starting from first principles, we derive the state-space model, implement the full prediction and update recursion, studying how the filter's uncertainty evolves over time under varied process and measurement noise configurations. The analysis then moves to an information-theoretic perspective through differential entropy and Kullback-Leibler divergence, connecting probabilistic confidence to information gain per observation. Results suggest that the balance between process and measurement noise governs both filter responsiveness and steady-state uncertainty, and that low uncertainty does not imply accuracy when process noise is underestimated.**

***Index Terms*—Kalman filter, Bayesian estimation, state-space model, uncertainty, differential entropy, KL divergence**

I. Introduction

In the domain of parameter-estimation, the "state" is the smallest set of variables denoting a system at work, with most problems involving a corrupting "noise" term that distorts measurement, analogous to a scenario involving global positioning systems (GPS), where their receiver's reported position will slightly differ from the true position of a moving vehicle by a random error each time they are sampled.

A computationally efficient solution is introduced by the Kalman filter. Through a mathematical model describing system dynamics, the algorithm finds the optimal estimate of the hidden state at each time step with a quantified measure of uncertainty by working recursively.

This report studies the Kalman filter applied to a 2-dimensional tracking problem in which an object moves with slowly varying velocity, observed through noisy position measurements only, with the following section breakdown:

- **Section II**: definition of the state-space model.
- **Section III**: examines a simulation of data.
- **Section IV**: presents the filter initialisation.
- **Section V**: analyses uncertainty evolution under varied noise.
- **Section VI**: provides an interpretation of results.

II. Model Definition

A. State Representation

The object's state at discrete time step k is the vector

$$x_k = \begin{bmatrix} p_x & v_x & p_y & v_y \end{bmatrix}^\top \in \mathbb{R}^4, \tag{1}$$

with interleaved position components p_x, p_y and velocity components v_x, v_y. The ordering format in (1) reflects the block-diagonal structure of the model matrices hereinbelow.

B. Dynamic Model

We adopt a constant-velocity motion model perturbed by small random accelerations:

$$x_k = F\, x_{k-1} + w_k, \qquad w_k \sim \mathcal{N}(0,\, Q). \tag{2}$$

The notation $w_k \sim \mathcal{N}(0, Q)$ means the noise w_k is drawn from a zero-mean Gaussian distribution with covariance matrix Q. In physical terms, the object follows constant-velocity kinematics on average, but experiences small unpredictable accelerations (modelled as Gaussian noise) causing fluctuations in the trajectory over time.

The transition matrix F propagates the state by one sampling period Δt under constant-velocity kinematics ($p_k = p_{k-1} + v_{k-1}\Delta t$, $v_k = v_{k-1}$):

$$F = \begin{bmatrix} 1 & \Delta t & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \Delta t \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{3}$$

C. Process Noise Covariance

The process noise covariance Q quantifies the magnitude of deviation from constant-velocity motion, derived by modelling the unknown acceleration as continuous white Gaussian noise with standard deviation σ_a, integrated over the sampling interval Δt. This yields correlated position and velocity perturbations:

$$Q = \sigma_a^2 \begin{bmatrix} \frac{\Delta t^4}{4} & \frac{\Delta t^3}{2} & 0 & 0 \\ \frac{\Delta t^3}{2} & \Delta t^2 & 0 & 0 \\ 0 & 0 & \frac{\Delta t^4}{4} & \frac{\Delta t^3}{2} \\ 0 & 0 & \frac{\Delta t^3}{2} & \Delta t^2 \end{bmatrix}. \tag{4}$$

The block-diagonal structure reflects independence between the x and y axes. Within each block, the off-diagonal entries are non-zero because a random acceleration perturbs velocity directly and position indirectly through integration; both of which are therefore correlated.

The scalar σ_a represents prior belief about the magnitude of unmodelled accelerations. In future sections, we use $\sigma_a = 1.5\ \mathrm{m/s^2}$ to produce visible trajectory curvature through accumulated drifts in velocity.

D. Measurement Model

The sensor inherently observes only position at each step, through which it infers velocity, and is modelled as:

$$z_k = H\,x_k + v_k, \qquad v_k \sim \mathcal{N}(0,\, R). \tag{5}$$

The measurement matrix H extracts position components from the full state:

$$H = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}. \tag{6}$$

The measurement noise covariance R is taken to be isotropic; equal, uncorrelated noise in both axes:

$$R = \sigma_z^2\, I_2, \tag{7}$$

where $\sigma_z = 1.0$ m is the position noise standard deviation and I_2 is the 2×2 identity. The isotropic form implies the sensor has no preferred axis of accuracy; as shown in Section IV, this is reflected in the circular uncertainty contours.

E. Initial State Distribution

Before any measurements are processed, the state is modelled as Gaussian:

$$x_0 \sim \mathcal{N}(m_{0|0},\, P_{0|0}). \tag{8}$$

where initial mean $m_{0|0}$ and covariance $P_{0|0}$ store prior belief about the object's initial state and our confidence thereof. The sensitivity of the filter of this design choice is examined in Section IV.

III. Simulation

A. Setup and Parameters

We simulate a ground-truth trajectory by propagating the dynamics of (2) forward from an initial state $x_0 = [0,\ 1,\ 0,\ \frac{1}{2}]^\top$ (position in m, velocity in m/s) over $T = 100$ steps at a sampling period of $\Delta t = 0.1$ s, giving a total duration of 10 s. At each step, process noise $w_k \sim \mathcal{N}(0, Q)$ is drawn and added to the next state. Noisy measurements are generated according to (5) by adding $v_k \sim \mathcal{N}(0, R)$ to the true position. The full parameter set is summarised in Table I.

TABLE I: Simulation Parameters

Parameter	Value	Description
Δt	0.1 s	Sampling period
T	100	Number of steps
σ_a	1.5 m/s^2	Process noise std dev
σ_z	1.0 m	Measurement noise std dev
x_0	$[10, 1, 8, 0.5]^\top$	Initial true state

B. Trajectory and Measurements

Fig. 1 shows the simulated ground-truth trajectory along with the noisy position measurements in the p_x-p_y plane. The naturally curving path is resultant from accumulated process noise causing the velocity to drift over time, consistent with $\sigma_a = 1.5$ m/s^2. The measurements scatter around the true positions with a spread consistent with $\sigma_z = 1.0$ m.

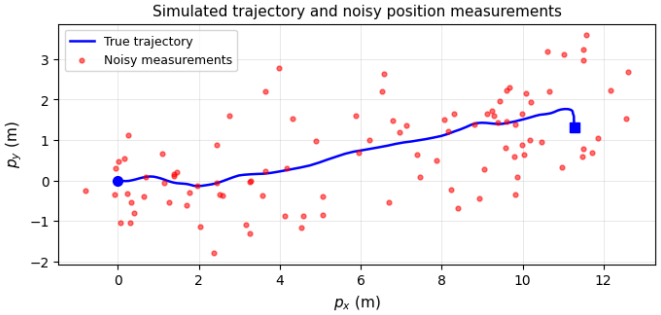


Fig. 1: Simulated 2D trajectory (blue) and noisy position measurements (red dots).

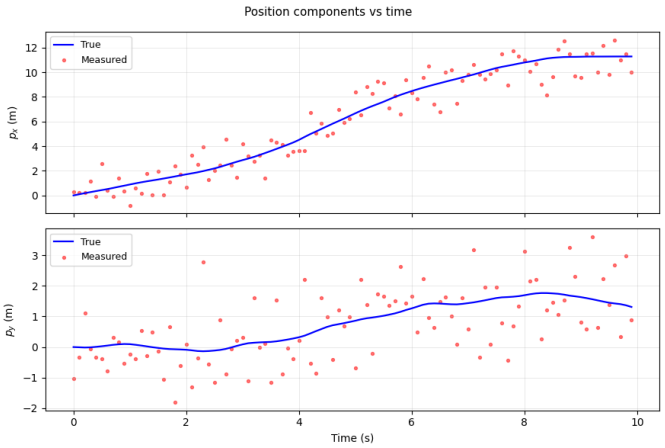


Fig. 2: Position components p_x (top) and p_y (bottom) vs time.

Fig. 2 shows the p_x and p_y components separately as functions of time. The simulations are now ready for the filtering task to be implemented and recover the smooth underlying signal from these corrupted observations, and in turn infer the unobserved velocity.

IV. Kalman Filter

A. The Kalman Recursion

The Kalman filter maintains a Gaussian belief over the hidden state at each time step, characterised by a mean vector $m_{k|k-1}$ and a covariance matrix representing the *predicted* distribution, $P_{k|k-1}$, and then refines both once measurement is received, calculating the *posterior* mean $m_{k|k}$ and covariance $P_{k|k}$ as shown in (9) and (10) respectively.

$$m_{k|k-1} = F\, m_{k-1|k-1}, \tag{9}$$
$$P_{k|k-1} = F\, P_{k-1|k-1}\, F^\top + Q. \tag{10}$$

When the measurement z_k arrives, the predicted distribution is corrected. The discrepancy between the actual measurement and what the model predicted is called the *innovation*:

$$r_k = z_k - H\, m_{k|k-1}. \tag{11}$$

The *innovation covariance* captures the total uncertainty in this discrepancy, combining prediction uncertainty with measurement noise:

$$S_k = H\, P_{k|k-1}\, H^\top + R. \tag{12}$$

The *Kalman gain* K_k is computed to determine how much weight to give the innovation relative to the prediction:

$$K_k = P_{k|k-1}\, H^\top S_k^{-1}. \tag{13}$$

A gain close to zero means the filter trusts the model prediction; a gain close to one means it trusts the measurement. The posterior is then:

$$m_{k|k} = m_{k|k-1} + K_k\, r_k, \tag{14}$$
$$P_{k|k} = (I_4 - K_k H)\, P_{k|k-1}. \tag{15}$$

In our implementation, K_k is computed by solving the linear system $S_k^\top K_k^\top = (P_{k|k-1}H^\top)^\top$ rather than inverting S_k explicitly, which is more numerically stable [2].

B. Initialisation

Two initialisations are compared. The first of which (zero-initialisation) sets $m_{0|0} = 0$ and $P_{0|0} = 10\, I_4$, this places the filter at the origin with uniform uncertainty ignoring where the object actually starts. Secondly (measurement initialisation), motivated by the implementation notes, we anchor position at the first noisy observation and set velocity to zero (no velocity data is available at $k = 0$):

$$m_{0|0} = \begin{bmatrix} z_{x,0} & 0 & z_{y,0} & 0 \end{bmatrix}^\top, \quad P_{0|0} = \mathrm{diag}(\sigma_z^2,\ \sigma_v^2,\ \sigma_z^2,\ \sigma_v^2), \tag{16}$$

where σ_z^2 inherits the known measurement noise level as the initial position uncertainty, and $\sigma_v^2 = 25\ \mathrm{m}^2/\mathrm{s}^2$ expresses ignorance of the initial velocity.

C. Results

Fig. 3 shows the filtered trajectory and 95% confidence ellipses at selected time steps for the measurement initialisation. The filtered estimate lies closer to the true path than the raw measurements because it makes use of the dynamic model to smoothen out measurement noise. The confidence ellipses are circular at all steps as a consequence of the isotropic $R = \sigma_z^2 I_2$ and the symmetric block structure of Q, which treats both position axes identically. The ellipses shrink rapidly from

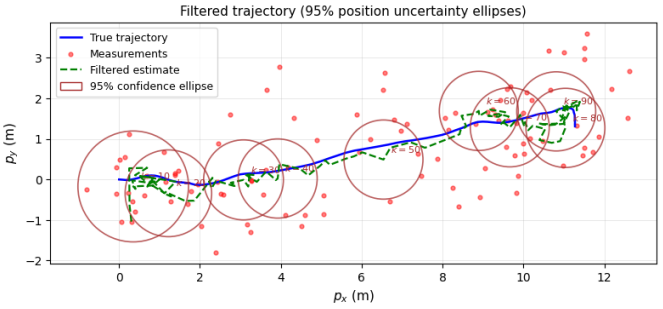


Fig. 3: Filtered trajectory (green dashes) against true path (blue) and measurements (red). Brown circles show 95% confidence ellipses at $k = 10-80$ in increments of 10. Ellipses are circular due to the isotropic sensor model.

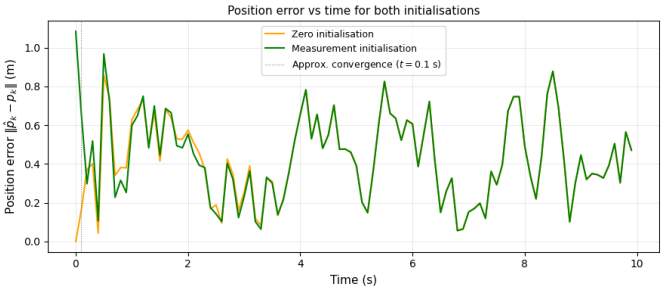


Fig. 4: Position error $\|\hat{p}_k - p_k\|$ vs time for zero initialisation (orange) and measurement initialisation (green).

$k = 10$ to $k = 20$, after which they become indistinguishable. This suggests that the filter reaches steady-state uncertainty within the simulation horizon.

Fig. 4 shows the position error $\|\hat{p}_k - p_k\|$ over time for both initialisations. Since the true trajectory starts at the origin and the zero initialisation also places the estimate there, both filters begin close to the truth. The measurement initialisation anchors approximately 1 m from the true starting position, thus incurring a marginally larger initial error. Both initialisations converge to the same error level within a single step: the high Kalman gain at $k = 1$, driven by the large initial covariance relative to R, causes the filter to almost fully weight the first measurement, correcting any residual offset early in simulation. After this point the two estimates are indistinguishable. Fig. 5 confirms that the filter recovers the unobserved velocity components despite H containing no velocity rows, with estimates tracking the true values after a short initial transient, albeit with some seeming phase shift.

V. Evolution of Uncertainty

To study how the Q/R balance affects filter behaviour, three noise arrangements are compared throughout this section and Section VI, varying the relative magnitudes of Q and R. **Case A** (σ_a=0.05 m/s^2, σ_z=2.0 m) assigns low process noise and high measurement noise, such that the filter strongly trusts its model over incoming observations; a small Q relative to R. **Case B** (σ_a=1.5 m/s^2, σ_z=1.0 m) is the balanced arrangement,

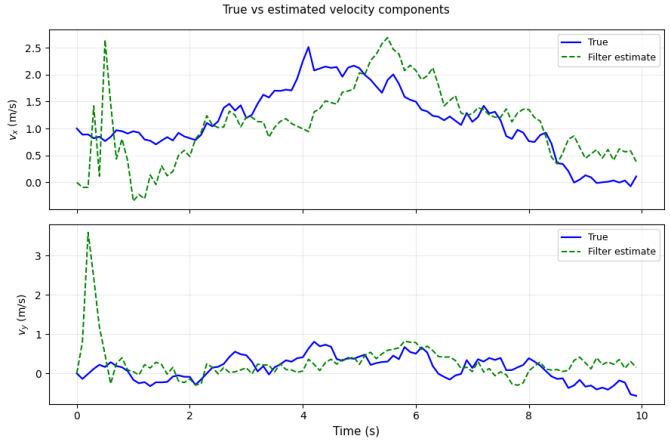


Fig. 5: Estimated (green dashed) vs true (blue) velocity components v_x and v_y. Velocity is never directly observed yet is recovered via the filter's coupled state representation.

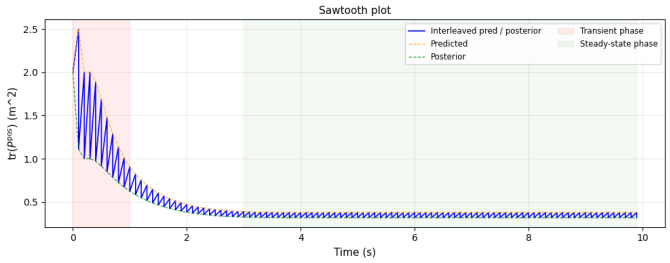


Fig. 6: Covariance trace sawtooth for Case B.

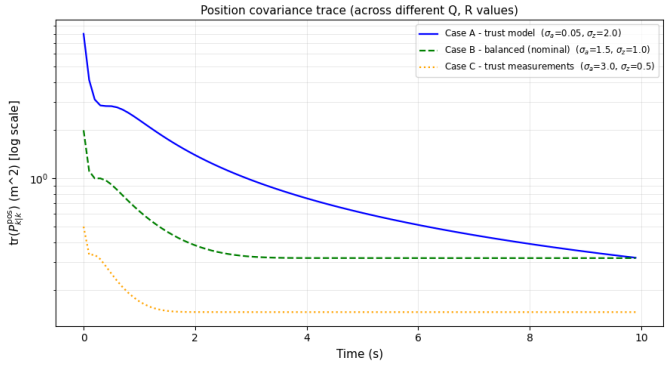


Fig. 7: Position covariance trace $\mathrm{tr}(P^{\mathrm{pos}}_{k|k})$ on a log scale for all three Q/R regimes. Case A reaches the lowest steady-state uncertainty; Case C the highest.

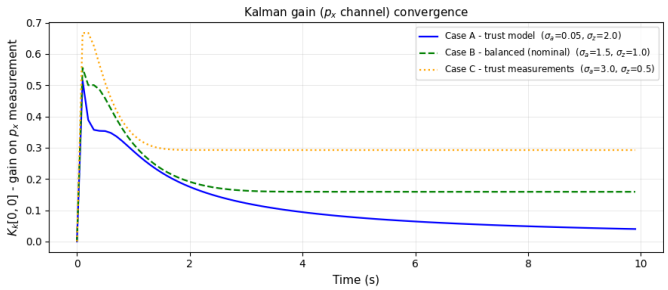


Fig. 8: Kalman gain $K_k[0,0]$ converging to steady-state values of 0.044, 0.159, and 0.293 for Cases A, B, and C respectively.

matching the simulation parameters from Section III. **Case C** (σ_a=3.0 m/s^2, σ_z=0.5 m) assigns high process noise and low measurement noise, so the filter strongly trusts measurements over its own prediction; a high Q relative to R.

A. The Predict-Update Cycle

A property of the Kalman filter is that it oscillates uncertainty at every step. During prediction, iterating through F and adding Q (in (10)) necessitates an increasing covariance, (i.e.: the object may have accelerated or decelerated unpredictably since the last measurement). Incorporating z_k during the update reduces covariance by the amount accrued in the Kalman gain. The net effect is a *sawtooth* pattern in the covariance trace.

Fig. 6 shows this for Case B. The transient phase (shaded red, $t < 1$ s) shows a steep initial descent as the large prior covariance reduces rapidly; the effect stemming from the first measurements. By $t \approx 3$ s (green shaded area) the filter reaches steady state, where uncertainty added by each prediction breaks even with the uncertainty removed by each update; a result of the discrete algebraic Riccati equation [2].

B. Effect of Q/R Ratio

Fig. 7 shows the covariance trace for all three cases on a log scale. Case A converges to the lowest steady-state uncertainty because it only injects a small process noise at each prediction. That is inherently observable because the filter maintains a tight belief by relying more on its own model than the other cases do. Case C converges to a higher steady-state value because each prediction is highly uncertain, requiring larger corrections from measurements. Case B lies between the two.

C. Kalman Gain Convergence

Fig. 8 shows the p_x channel gain $K_k[0,0]$ over time. All three cases begin with a high gain. When the prior covariance is large, the filter assigns high weight to measurements. As covariance reduces, the gain reduces to a case-specific steady-state: 0.044 for Case A, 0.159 for Case B, and 0.293 for Case C. A gain of 0.044 means only $\approx 4\%$ of each innovation is used to correct the estimate, with the remaining $\approx 96\%$ coming from the model prediction. A gain of 0.293 means $\approx 29\%$ of each correction comes from the measurement. This is the quantitative expression of the trust balance impacted by the Q/R ratio.

VI. Information-Theoretic Analysis

A. Definitions

Two quantities from information theory are used to derive filter characteristics. Throughout this section all logarithms are natural, so quantities are expressed in *nats* (the unit of information when using base-e logarithms).

The differential entropy of a Gaussian $\mathcal{N}(m, P)$ over $\mathbb{R}^4$ measures the spread of the distribution:

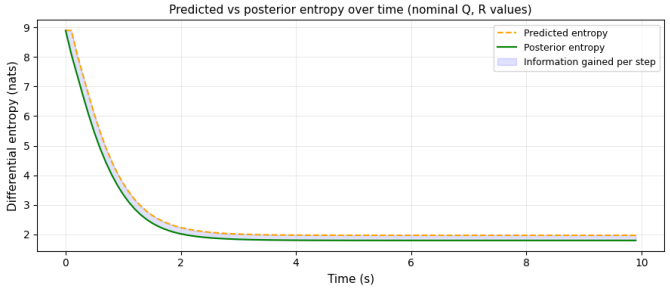


Fig. 9: Predicted (orange dashed) and posterior (green) differential entropy for Case B. The shaded gap represents information gained at each step.

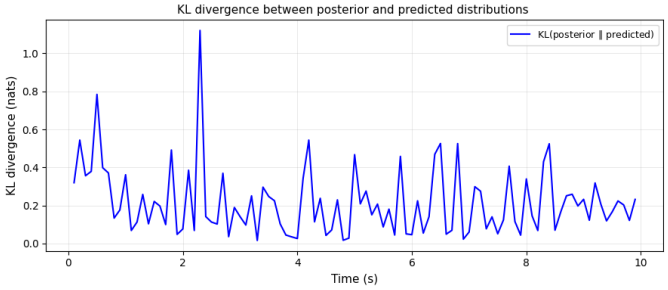


Fig. 10: KL divergence between posterior and predicted distributions for Case B.

$$h(P) = \tfrac{1}{2}\log\big[(2\pi e)^4 \det(P)\big]\,. \tag{17}$$

A small or negative h implies a concentrated distribution; a large or positive h indicates a diffuse, uncertain one.

The Kullback-Leibler (KL) divergence from the posterior $\mathcal{N}(m_{k|k}, P_{k|k})$ to the predicted $\mathcal{N}(m_{k|k-1}, P_{k|k-1})$ quantifies how much the measurement shifted the distribution at each step:

$$D_{\mathrm{KL}} = \tfrac{1}{2}\left[\log\frac{\det P_{k|k-1}}{\det P_{k|k}} - 4 + \mathrm{tr}(P_{k|k-1}^{-1}P_{k|k}) + \delta^\top P_{k|k-1}^{-1}\delta\right], \tag{18}$$

where $\delta = m_{k|k-1} - m_{k|k}$. Both log-determinant terms are computed via `slogdet` for numerical stability, and matrix inverses are evaluated via linear solves rather than explicit inversion [2].

B. Entropy Reduction as Information Gain

Fig. 9 shows predicted and posterior entropy over time for Case B. The posterior is always below the predicted at every step, each measurement reducing uncertainty. The shaded gap between the curves represents information gained per observation at each step. Contrary to what can visually be seen from the figure alone, this gap is largest during the transient phase, 0.811 nats at $k = 1$ because the prior covariance is broad and each measurement produces a large correction. It follows up by reducing to a steady-state mean of 0.173 nats once the covariance has converged. This suggests a fixed rate of information gain per observation.

Fig. 10 shows the KL divergence between the posterior and predicted distributions at each step for Case B. The KL divergence fluctuates throughout the entire run including steady state, possibly because KL divergence depends on both covariance ratio (early stabilisation) and the quadratic term $\delta^\top P_{k|k-1}^{-1}\delta$ where $\delta = m_{k|k-1} - m_{k|k}$ is the mean shift induced by the measurement. This term is driven by the actual innovation $r_k = z_k - Hm_{k|k-1}$ at each step, and this parameter varies stochastically regardless of the covariance's settling. A large KL at a given step therefore indicates that the measurement was particularly surprising relative to the prediction at that moment. The steady-state mean of approximately 0.22 nats reflects the average information gained per observation once the covariance has converged.

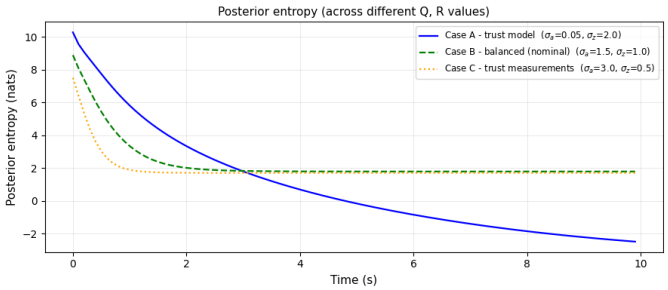


Fig. 11: Steady-state posterior entropy: Case A (-2.184 nats), Case B (1.795 nats), Case C (1.709 nats). Case A's negative entropy reflects an extremely concentrated distribution.

C. Dependence on Noise Levels

Figs. 11 and 12 compare both quantities across all three cases. Case A reaches the lowest steady-state posterior entropy (-2.184 nats) showing that the filter is highly concentrated and confident. Cases B and C stabilise at 1.795 and 1.709 nats respectively, indicating considerably more residual uncertainty.

The KL divergence in Fig. 12 tells the complementary story. Case C reaches the highest steady-state KL (1.521 nats), due to its prediction being highly corrected by each precise measurement, so each observation is supplementary informative. Case A has the lowest KL (0.022 nats), its tight prediction already anticipates where the measurement will land, so almost nothing is learned at each step.

D. Overconfidence and Its Consequences

The most important observation connecting Sections V and VI is that low entropy does not imply accuracy. Case A is the most concentrated filter, yet achieves this by underestimating process noise ($\sigma_a = 0.05$), causing it to discount measurements in favour of its own model. A filter that learns little from each observation ($K_k[0,0] = 0.044$, $D_{\mathrm{KL}} = 0.022$ nats) is slow to correct errors introduced by unmodelled dynamics. The analysis of σ_a choices reveals that setting it too small produces a confident but poorly adapted filter.

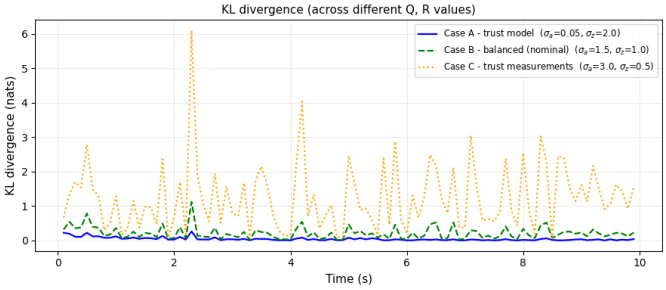


Fig. 12: Steady-state KL divergence: Case A (0.022 nats), Case B (0.220 nats), Case C (1.521 nats). Higher KL indicates each measurement is more informative relative to the prediction.

VII. Conclusion

This report has implemented and analysed a 2D constant-velocity Kalman filter from first principles. The filter recovers both position and unobserved velocity from noisy measurements, with uncertainty undergoing the characteristic predict-update sawtooth before converging to a steady state governed by the Q/R ratio. Information-theoretic experiments confirm that differential entropy and KL divergence quantify the same convergence behaviour in units of information, and reveal that a filter can be simultaneously confident and poorly calibrated when process noise is underestimated. Together, the uncertainty and information analyses demonstrate that principled noise parameter selection is as important as correct implementation of the recursion itself.

References

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[2] S. Särkkä, *Bayesian Filtering and Smoothing*. Cambridge University Press, 2013.